

DMYTRO VOLOSHYN

**Software & Solutions Architect | Product Owner | Java Applications Engineer
(Desktop & Android) | AI & Integrated Systems Expert**

Location: Sitia, Crete, Greece

Availability: Remote collaboration globally

Email: dmvsense@gmail.com **Phone:** +30 697 658 3857

PROFESSIONAL LINKS

LinkedIn — Professional Profile: <https://www.linkedin.com/in/dmytro-voloshyn-software-architect/>

GitHub — Engineering Patterns: <https://github.com/dmvoloshyn>

GitHub — Software Projects: <https://github.com/DmV-Software>

DmV Software — Commercial Products: <https://dmvtools.gumroad.com/>

PROFESSIONAL SUMMARY

Software & Solutions Architect and Product Owner specializing in the end-to-end development of commercial B2B applications, scientific software, engineering utilities, and AI-driven solutions.

I combine modern software engineering with 30+ years of multidisciplinary experience in industrial electronics, embedded systems, digital control, instrumentation, industrial automation, and hardware-software integration. This systems-engineering background provides a practical understanding of physical constraints, reliability, measurement, performance, and real-world system behavior that directly informs my current software architecture and engineering decisions.

I design and develop complete applications from domain analysis and algorithmic modeling through architecture, calculation engines, UI/UX logic, data modeling and persistence, database integration, testing, packaging, and commercial release. My primary software focus is Java, JavaFX desktop applications, and Android applications, with additional expertise in C/C++ integration through JNI and AI/LLM-based application architectures.

Recent work includes scientific marine-data software integrating native C++ libraries, an independent multi-domain AI research agent orchestrating multiple NASA APIs using Java, LangChain4j and Gemini, an 8-language commercial construction calculation application, and an offline-first Android engineering utility.

My approach is focused on robust, maintainable, performance-oriented software and complete product ownership, with AI used as an engineering accelerator while retaining full technical control over architecture, code quality, validation, and final product behavior.

CORE COMPETENCIES

Software Architecture & Product Development

- Solutions Architecture & System Architecture
- End-to-End Product Ownership
- Domain Modeling & Algorithmic Design
- Application Architecture & Modular Design
- Performance Optimization
- UI/UX Logic & Technical Workflow Design
- Requirements Analysis & Technical Feasibility
- Software Modernization & Legacy System Modernization
- Commercial Packaging and Product Release

Java & Application Engineering

- Java Application Development
- JavaFX Desktop Architecture
- Android Development
- Complex Engineering Calculation Engines
- Persistent Application State & History Management
- Database Integration
- PDF Generation and Automated Document Workflows
- Multilingual and Multi-Currency Application Architecture

AI & Advanced Integrations

- LLM Integration
- Agentic AI Workflows
- LangChain4j
- Gemini
- Multi-Step Tool Chaining
- REST API Orchestration
- RAG & Live-Data Grounding
- AI-Assisted Development, Refactoring & Testing
- Resilient API Integration and Fallback Strategies

Scientific & Systems Software

- Scientific Data Processing

- C/C++ Integration through JNI
- GeoJSON and Spatial Data Processing
- High-Performance Computational Workflows
- Engineering Data Validation and Quality Control

Systems Engineering & R&D

- 30+ Years Industrial Engineering Experience
 - Embedded Systems
 - Microprocessor Programming — Assembly
 - Industrial Automation & Digital Control
 - Power Electronics
 - Precision Measurement & Instrumentation
 - PCB and Electronic System Design
 - Hardware-Software Integration
 - Technology Readiness Level (TRL) Assessment
 - Technical Feasibility Studies
 - R&D Strategy and Multidisciplinary Technical Leadership
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PROFESSIONAL EXPERIENCE

Independent Software & Solutions Architect | Product Owner

DmV Software | 2020 – Present

Sitia, Crete, Greece

Independent architect, lead developer and product owner responsible for the complete lifecycle of commercial, engineering, scientific and AI-driven software products.

Software Product Development

- Architect and develop complete software products from initial concept and domain analysis through implementation, validation, packaging and commercial release.
- Design robust, modular and maintainable Java application architectures with emphasis on reliability, performance and practical usability.
- Develop engineering applications containing complex parametric calculation engines, persistent state management, multilingual interfaces, multi-currency support and automated document generation.
- Own product decisions covering functionality, application architecture, UI/UX logic, technical implementation, testing and commercial deployment.

Scientific & Marine Software — Marine Data Lab

- Led the end-to-end development of a scientific desktop application for processing and analyzing large-scale oceanographic datasets.
- Integrated high-performance native C++ scientific libraries into the Java environment using JNI (Java Native Interface) for computationally intensive workflows.
- Implemented spatial data processing using GeoJSON libraries and applied structured quality-control procedures to regional ocean-model data.
- Integrated application data and database components to support reliable processing and management of complex scientific datasets.

Agentic AI — NASA Enterprise Mission Control Agent

- Architected an independent multi-domain AI research agent using Java, LangChain4j and Gemini.
- Engineered multi-step tool chaining to orchestrate 10 fragmented NASA APIs, with RAG-based knowledge grounding through a natural-language interface.
- Implemented resilient API fallback mechanisms and strict live-data grounding principles to improve reliability and reduce hallucination risk.
- Designed the system as a controlled agentic workflow in which LLM reasoning is combined with deterministic external tools and verified live data.

B2B Commercial Software — Concrete Works Cost Estimator Pro

- Developed and commercially packaged an 8-language Windows desktop application using Java/JavaFX for automated construction calculations and PDF-based quoting.
- Designed a dual-mode calculation engine supporting dynamic and customizable engineering coefficients.
- Implemented an advanced History Journal providing persistent state management, intelligent autofill, cross-session data transfer and search capabilities.
- Implemented localized EU-oriented formatting, multilingual workflows and multi-currency support.
- Managed the complete product lifecycle from domain modeling and architecture through development, testing, packaging and commercial distribution.

Android Engineering Software — DmV Calculator Pro

- Designed and developed a high-precision Android engineering utility using Java.
 - Implemented complex unit transformations and technical formula processing with an offline-first architecture.
 - Designed persistent operation history and field-oriented workflows for practical use without dependence on continuous connectivity.
 - Developed the product as a complete end-user application, including UI/UX, calculation logic, testing and commercial packaging.
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SENIOR SYSTEMS ENGINEERING & R&D LEADERSHIP

Industrial Electronics, Embedded Systems & Automation | Earlier Career

30+ years of engineering experience covering industrial electronics, embedded systems, digital control, instrumentation, power electronics, automation and multidisciplinary R&D.

R&D and Technical Leadership

- Directed R&D radio-electronics laboratories and led multidisciplinary engineering teams developing systems for production and commercial applications.
- Managed hardware/software integration strategies and architectural decisions required to transform functional prototypes into reliable, scalable systems.
- Conducted technical feasibility studies, technology evaluations, implementation-readiness assessments and Technology Readiness Level (TRL) evaluations.
- Evaluated engineering solutions from both technical and practical production perspectives, including reliability, manufacturability and system integration.

Embedded Systems & Low-Level Architecture

- Developed microprocessor-based control systems using Assembly, translating strict hardware and timing constraints into optimized software execution.
- Designed embedded control architectures and interfaces for industrial and specialized technical systems.
- Applied low-level understanding of processors, electronics, measurement and control to system-level engineering decisions.

Industrial Electronics & Automation

- Executed end-to-end electronic system design, including circuit development, PCB design and fabrication supervision.
- Designed digital control systems for industrial conveyor installations and automated equipment.
- Developed precision measurement and monitoring systems for demanding industrial environments.
- Worked across power electronics, digital control, instrumentation and industrial automation.

Marine & Specialized Systems

- Developed a microprocessor-based insulation-resistance monitoring system for Remotely Operated Underwater Vehicles (ROV), addressing critical electrical-system integrity requirements.
- Designed systems combining embedded processing, electrical measurement, control logic and specialized marine hardware constraints.

- Applied systems-engineering principles to the integration of electronics, software, measurement and physical equipment.
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SELECTED SOFTWARE, AI & ENGINEERING PROJECTS

Marine Data Lab

Java | C++ | JNI | GeoJSON | Scientific Data Processing

Scientific desktop application for large-scale oceanographic data processing and analysis, integrating native C++ scientific libraries into Java and supporting spatial data quality-control workflows.

NASA Enterprise Mission Control Agent

Java | LangChain4j | Gemini | REST APIs | Agentic AI

Independent multi-domain research agent orchestrating 10 NASA APIs through natural-language interaction, combining LLM reasoning with multi-step tool chaining, live-data grounding and resilient API integration.

Concrete Works Cost Estimator Pro

Java | JavaFX | Desktop | Engineering Calculations | PDF | History | 8 Languages

Commercial B2B engineering application for concrete and reinforced-concrete cost estimation, featuring parametric/manual calculation modes, persistent History Journal, multilingual localization, multi-currency support and automated PDF quotations.

DmV Calculator Pro

Java | Android | Offline-First | Engineering Calculations & Conversions

Mobile engineering utility for technical formulas and unit transformations, featuring persistent operation history and a field-oriented user interface.

ROV Insulation Resistance Monitoring System

Embedded Systems | Microprocessors | Assembly | Electrical Measurement

Microprocessor-based monitoring system for insulation resistance and electrical-system integrity in Remotely Operated Underwater Vehicles.

Industrial Conveyor Control Systems

Digital Control | Industrial Automation | Embedded Systems

Digital control complexes for automated conveyor installations, integrating control logic, electronic hardware and system reliability requirements.

Industrial Power Electronics

Power Electronics | Digital Control | Precision Measurement

Engineering methodologies and electronic systems for specialized power-supply and transformer applications, including high-current and microcurrent precision applications.

EDUCATION

Master's Degree (Specialist) — Electrical Engineering and Electromechanics

Specialization: Industrial Electronics

National University of Shipbuilding (NUOS)

Integrated 5.5-year program | 1982–1988

LANGUAGES

English: Professional Working Proficiency (B2)

Spanish: Basic (A2)

Ukrainian / Russian: Native

ADDITIONAL INFORMATION

Work Authorization: Eligible to work in Greece

Availability: Remote collaboration globally

Target Areas: Software & Solutions Architecture, Java Desktop Applications, Android Engineering Applications, Scientific Software, AI/LLM Integration, Industrial & Marine Technology, Technical Consulting, R&D and Hardware-Software Integrated Systems.